

# MA374 Financial Engineering Lab 07

## Exploratory Analysis of Stock Returns

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### 1 Introduction

This report presents an exploratory statistical analysis of stock return data from the National Stock Exchange (NSE) and Bombay Stock Exchange (BSE). The objective is to investigate whether the empirical behaviour of stock returns aligns with the assumptions of classical financial models such as the Black–Scholes framework.

The analysis includes:

- Descriptive statistics
- Boxplots and quantile–quantile (QQ) plots
- Normality tests (Kolmogorov–Smirnov and Shapiro–Wilk)
- Maximum likelihood estimation of parameters

The study is conducted for daily returns, log returns, weekly returns, and monthly returns.

### 2 Daily Returns Analysis

Daily returns are computed using

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

where  $P_t$  denotes the price at time  $t$ .

## 2.1 Descriptive Statistics

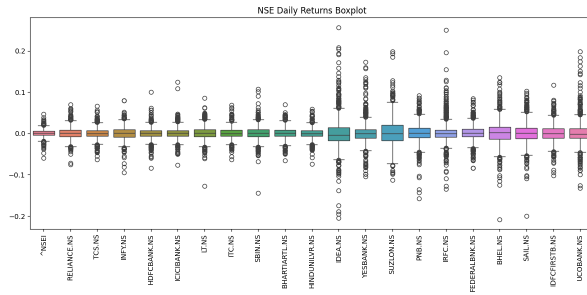
### 2.1.1 NSE

|               | Mean     | Std Dev  | Skewness  | Kurtosis  |
|---------------|----------|----------|-----------|-----------|
| $^N SEI$      | 0.000568 | 0.008786 | -0.361435 | 4.283443  |
| RELIANCE.NS   | 0.000604 | 0.014173 | 0.196660  | 2.809449  |
| TCS.NS        | 0.000211 | 0.012992 | 0.127499  | 2.562334  |
| INFY.NS       | 0.000440 | 0.015067 | -0.186322 | 3.838863  |
| HDFCBANK.NS   | 0.000425 | 0.013354 | 0.081990  | 5.756245  |
| ICICIBANK.NS  | 0.000877 | 0.013879 | 0.826138  | 9.424728  |
| LT.NS         | 0.001073 | 0.015200 | -0.214954 | 5.850605  |
| ITC.NS        | 0.000838 | 0.012522 | 0.323128  | 3.327425  |
| SBIN.NS       | 0.001222 | 0.016460 | 0.008638  | 8.669389  |
| BHARTIARTL.NS | 0.001232 | 0.014190 | 0.161916  | 1.751723  |
| HINDUNILVR.NS | 0.000158 | 0.012632 | 0.094496  | 3.225802  |
| IDEA.NS       | 0.000750 | 0.037422 | 0.814931  | 7.377200  |
| YESBANK.NS    | 0.000555 | 0.024937 | 1.460684  | 9.183042  |
| SUZLON.NS     | 0.002352 | 0.033266 | 0.797591  | 2.837920  |
| PNB.NS        | 0.001396 | 0.022878 | -0.522271 | 5.526291  |
| IRFC.NS       | 0.001795 | 0.025773 | 1.771503  | 14.364211 |
| FEDERALBNK.NS | 0.001278 | 0.018084 | 0.110297  | 2.847928  |
| BHEL.NS       | 0.002099 | 0.027945 | -0.073607 | 4.904415  |
| SAIL.NS       | 0.001210 | 0.025982 | -0.178048 | 4.273494  |
| IDFCFIRSTB.NS | 0.000689 | 0.020237 | 0.207915  | 2.997264  |
| UCOBANK.NS    | 0.001047 | 0.027405 | 1.421635  | 9.610414  |

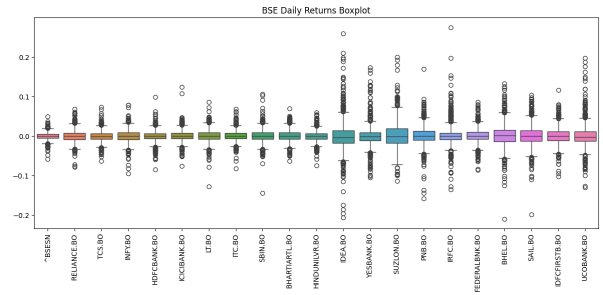
### 2.1.2 BSE

|               | Mean     | Std Dev  | Skewness  | Kurtosis  |
|---------------|----------|----------|-----------|-----------|
| ĤSESN         | 0.000546 | 0.008835 | -0.282446 | 4.009063  |
| RELIANCE.BO   | 0.000561 | 0.014437 | 0.040294  | 3.154887  |
| TCS.BO        | 0.000209 | 0.013148 | 0.234741  | 2.937532  |
| INFY.BO       | 0.000451 | 0.015264 | -0.139171 | 3.977793  |
| HDFCBANK.BO   | 0.000478 | 0.013379 | 0.069213  | 5.769860  |
| ICICIBANK.BO  | 0.000894 | 0.013980 | 0.833148  | 9.194905  |
| LT.BO         | 0.001085 | 0.015307 | -0.297107 | 6.349477  |
| ITC.BO        | 0.000800 | 0.012678 | 0.143895  | 4.469119  |
| SBIN.BO       | 0.001253 | 0.016556 | 0.012185  | 8.638470  |
| BHARTIARTL.BO | 0.001242 | 0.014293 | 0.184591  | 1.684581  |
| HINDUNILVR.BO | 0.000150 | 0.012600 | 0.100454  | 3.349110  |
| IDEA.BO       | 0.000750 | 0.037437 | 0.829120  | 7.510431  |
| YESBANK.BO    | 0.000569 | 0.025309 | 1.519434  | 9.398297  |
| SUZLON.BO     | 0.002341 | 0.033995 | 0.814074  | 2.986148  |
| PNB.BO        | 0.001429 | 0.023389 | -0.198271 | 7.174726  |
| IRFC.BO       | 0.001830 | 0.026093 | 1.973114  | 17.363667 |
| FEDERALBNK.BO | 0.001301 | 0.018134 | 0.165995  | 2.834014  |
| BHEL.BO       | 0.002139 | 0.028119 | -0.079223 | 4.873259  |
| SAIL.BO       | 0.001244 | 0.026210 | -0.131911 | 4.173894  |
| IDFCFIRSTB.BO | 0.000698 | 0.020192 | 0.222079  | 3.042781  |
| UCOBANK.BO    | 0.001068 | 0.027576 | 1.423095  | 9.568927  |

## 2.2 Daily Return Boxplots



(a) NSE Daily Returns

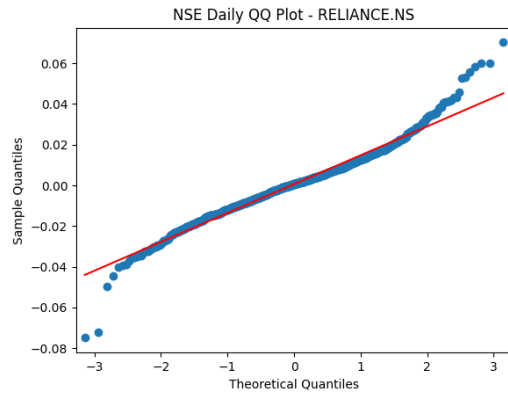


(b) BSE Daily Returns

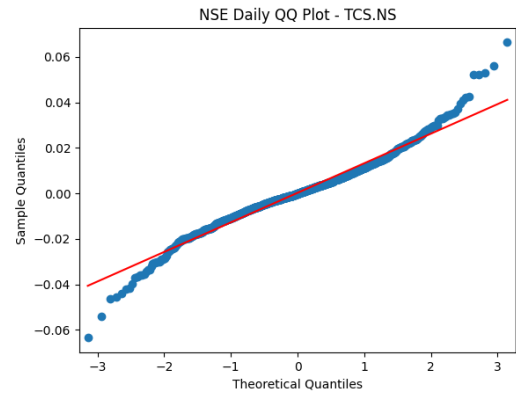
Figure 1: Boxplots of daily returns

**Observation:** The boxplots show several outliers, indicating large fluctuations in stock prices.

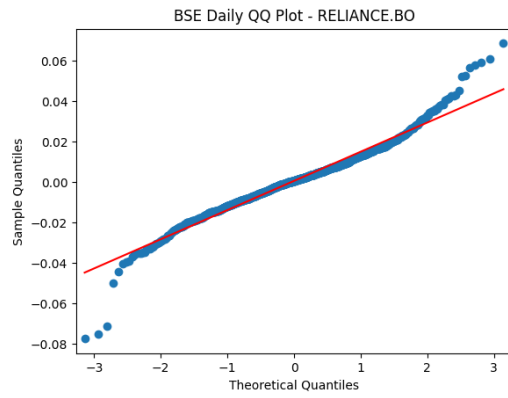
## 2.3 Daily Return QQ Plots



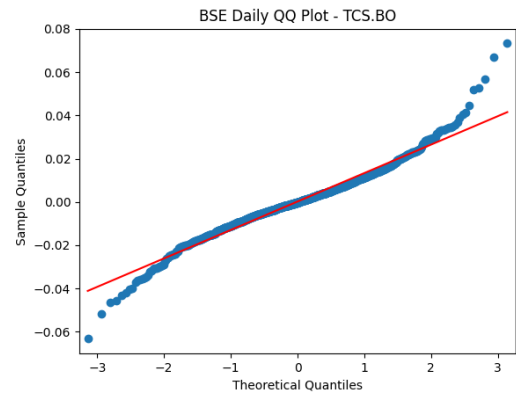
(a) RELIANCE (NSE)



(b) TCS (NSE)



(c) RELIANCE (BSE)



(d) TCS (BSE)

Figure 2: Representative QQ plots for daily returns

**Observation:** The tails deviate from the reference line, indicating heavy-tailed distributions.

## 2.4 Normality Tests

### 2.4.1 Kolmogorov–Smirnov Test

|               | KS Statistic | p-value  |
|---------------|--------------|----------|
| $^N SEI$      | 0.055587     | 0.001053 |
| RELIANCE.NS   | 0.056977     | 0.000719 |
| TCS.NS        | 0.055034     | 0.001222 |
| INFY.NS       | 0.045887     | 0.011614 |
| HDFCBANK.NS   | 0.067134     | 0.000033 |
| ICICIBANK.NS  | 0.070123     | 0.000012 |
| LT.NS         | 0.054792     | 0.001304 |
| ITC.NS        | 0.057707     | 0.000586 |
| SBIN.NS       | 0.072542     | 0.000005 |
| BHARTIARTL.NS | 0.053360     | 0.001903 |
| HINDUNILVR.NS | 0.062141     | 0.000160 |
| IDEA.NS       | 0.113000     | 0.000000 |
| YESBANK.NS    | 0.119680     | 0.000000 |
| SUZLON.NS     | 0.076336     | 0.000001 |
| PNB.NS        | 0.065835     | 0.000051 |
| IRFC.NS       | 0.158154     | 0.000000 |
| FEDERALBNK.NS | 0.065476     | 0.000057 |
| BHEL.NS       | 0.064142     | 0.000087 |
| SAIL.NS       | 0.073119     | 0.000004 |
| IDFCFIRSTB.NS | 0.056333     | 0.000859 |
| UCOBANK.NS    | 0.112381     | 0.000000 |

### 2.4.2 Shapiro–Wilk Test

|               | Shapiro Statistic | p-value  |
|---------------|-------------------|----------|
| $N_{SEI}$     | 0.959419          | 0.000000 |
| RELIANCE.NS   | 0.969608          | 0.000000 |
| TCS.NS        | 0.970509          | 0.000000 |
| INFY.NS       | 0.964192          | 0.000000 |
| HDFCBANK.NS   | 0.943769          | 0.000000 |
| ICICIBANK.NS  | 0.927130          | 0.000000 |
| LT.NS         | 0.953892          | 0.000000 |
| ITC.NS        | 0.961549          | 0.000000 |
| SBIN.NS       | 0.930614          | 0.000000 |
| BHARTIARTL.NS | 0.979191          | 0.000000 |
| HINDUNILVR.NS | 0.961591          | 0.000000 |
| IDEA.NS       | 0.892756          | 0.000000 |
| YESBANK.NS    | 0.862256          | 0.000000 |
| SUZLON.NS     | 0.959524          | 0.000000 |
| PNB.NS        | 0.943114          | 0.000000 |
| IRFC.NS       | 0.821224          | 0.000000 |
| FEDERALBNK.NS | 0.961028          | 0.000000 |
| BHEL.NS       | 0.951536          | 0.000000 |
| SAIL.NS       | 0.959872          | 0.000000 |
| IDFCFIRSTB.NS | 0.965883          | 0.000000 |
| UCOBANK.NS    | 0.866322          | 0.000000 |

## 2.5 MLE Estimates

|               | Mean (MLE) | Variance (MLE) | CI Lower  | CI Upper |
|---------------|------------|----------------|-----------|----------|
| $^N SEI$      | 0.000568   | 0.000077       | 0.000074  | 0.001062 |
| RELIANCE.NS   | 0.000604   | 0.000201       | -0.000193 | 0.001401 |
| TCS.NS        | 0.000211   | 0.000169       | -0.000520 | 0.000941 |
| INFY.NS       | 0.000440   | 0.000227       | -0.000407 | 0.001287 |
| HDFCBANK.NS   | 0.000425   | 0.000178       | -0.000325 | 0.001176 |
| ICICIBANK.NS  | 0.000877   | 0.000193       | 0.000097  | 0.001658 |
| LT.NS         | 0.001073   | 0.000231       | 0.000219  | 0.001928 |
| ITC.NS        | 0.000838   | 0.000157       | 0.000134  | 0.001542 |
| SBIN.NS       | 0.001222   | 0.000271       | 0.000296  | 0.002147 |
| BHARTIARTL.NS | 0.001232   | 0.000201       | 0.000435  | 0.002030 |
| HINDUNILVR.NS | 0.000158   | 0.000160       | -0.000552 | 0.000868 |
| IDEA.NS       | 0.000750   | 0.001400       | -0.001354 | 0.002855 |
| YESBANK.NS    | 0.000555   | 0.000622       | -0.000847 | 0.001957 |
| SUZLON.NS     | 0.002352   | 0.001107       | 0.000482  | 0.004223 |
| PNB.NS        | 0.001396   | 0.000523       | 0.000110  | 0.002683 |
| IRFC.NS       | 0.001795   | 0.000664       | 0.000346  | 0.003244 |
| FEDERALBNK.NS | 0.001278   | 0.000327       | 0.000261  | 0.002295 |
| BHEL.NS       | 0.002099   | 0.000781       | 0.000527  | 0.003670 |
| SAIL.NS       | 0.001210   | 0.000675       | -0.000251 | 0.002671 |
| IDFCFIRSTB.NS | 0.000689   | 0.000410       | -0.000448 | 0.001827 |
| UCOBANK.NS    | 0.001047   | 0.000751       | -0.000494 | 0.002588 |

## 3 Log Returns

Log returns are defined as

$$r_t = \ln \left( \frac{P_t}{P_{t-1}} \right)$$

Log returns are commonly used in financial modelling due to their time-additive properties.

### 3.1 Descriptive Statistics

#### 3.1.1 NSE

|               | Mean     | Standard Deviation | Skewness  | Kurtosis  |
|---------------|----------|--------------------|-----------|-----------|
| $^N SEI$      | 0.000529 | 0.008797           | -0.443730 | 4.502435  |
| RELIANCE.NS   | 0.000503 | 0.014152           | 0.094678  | 2.846177  |
| TCS.NS        | 0.000126 | 0.012984           | 0.039051  | 2.546115  |
| INFY.NS       | 0.000326 | 0.015091           | -0.320098 | 4.103849  |
| HDFCBANK.NS   | 0.000336 | 0.013351           | -0.071106 | 5.598013  |
| ICICIBANK.NS  | 0.000782 | 0.013801           | 0.615518  | 8.192156  |
| LT.NS         | 0.000957 | 0.015223           | -0.404787 | 6.891390  |
| ITC.NS        | 0.000759 | 0.012492           | 0.225571  | 3.253711  |
| SBIN.NS       | 0.001086 | 0.016463           | -0.269362 | 9.838979  |
| BHARTIARTL.NS | 0.001131 | 0.014162           | 0.082469  | 1.762325  |
| HINDUNILVR.NS | 0.000078 | 0.012628           | -0.005290 | 3.313712  |
| IDEA.NS       | 0.000061 | 0.037055           | 0.322367  | 7.184049  |
| YESBANK.NS    | 0.000251 | 0.024545           | 1.136446  | 7.956804  |
| SUZLON.NS     | 0.001807 | 0.032832           | 0.603159  | 2.158849  |
| PNB.NS        | 0.001132 | 0.023031           | -0.791768 | 6.869602  |
| IRFC.NS       | 0.001471 | 0.025255           | 1.298494  | 11.770059 |
| FEDERALBNK.NS | 0.001114 | 0.018058           | -0.021162 | 2.875848  |
| BHEL.NS       | 0.001706 | 0.027997           | -0.389009 | 6.198611  |
| SAIL.NS       | 0.000871 | 0.026072           | -0.449777 | 5.749680  |
| IDFCFIRSTB.NS | 0.000485 | 0.020202           | 0.056998  | 3.017693  |
| UCOBANK.NS    | 0.000680 | 0.026947           | 1.046092  | 8.223287  |

### 3.1.2 BSE

|               | Mean     | Standard Deviation | Skewness  | Kurtosis  |
|---------------|----------|--------------------|-----------|-----------|
| $BSES_N$      | 0.000507 | 0.008844           | -0.361944 | 4.177988  |
| RELIANCE.BO   | 0.000457 | 0.014433           | -0.072854 | 3.294457  |
| TCS.BO        | 0.000123 | 0.013131           | 0.139367  | 2.840547  |
| INFY.BO       | 0.000334 | 0.015285           | -0.277777 | 4.201794  |
| HDFCBANK.BO   | 0.000389 | 0.013376           | -0.084645 | 5.629244  |
| ICICIBANK.BO  | 0.000797 | 0.013900           | 0.626013  | 7.965843  |
| LT.BO         | 0.000967 | 0.015341           | -0.498848 | 7.427029  |
| ITC.BO        | 0.000720 | 0.012663           | 0.019657  | 4.621783  |
| SBIN.BO       | 0.001115 | 0.016558           | -0.266524 | 9.798514  |
| BHARTIARTL.BO | 0.001139 | 0.014262           | 0.106283  | 1.683890  |
| HINDUNILVR.BO | 0.000071 | 0.012596           | -0.001409 | 3.437699  |
| IDEA.BO       | 0.000060 | 0.037063           | 0.331276  | 7.271883  |
| YESBANK.BO    | 0.000256 | 0.024888           | 1.189179  | 8.118987  |
| SUZLON.BO     | 0.001773 | 0.033539           | 0.608830  | 2.304037  |
| PNB.BO        | 0.001153 | 0.023470           | -0.528824 | 7.927954  |
| IRFC.BO       | 0.001499 | 0.025513           | 1.426293  | 13.656158 |
| FEDERALBNK.BO | 0.001136 | 0.018098           | 0.035222  | 2.836047  |
| BHEL.BO       | 0.001742 | 0.028172           | -0.396124 | 6.209233  |
| SAIL.BO       | 0.000900 | 0.026284           | -0.400943 | 5.551260  |
| IDFCFIRSTB.BO | 0.000495 | 0.020155           | 0.070390  | 3.055715  |
| UCOBANK.BO    | 0.000696 | 0.027111           | 1.048142  | 8.139145  |

### 3.2 Daily vs Log Return Distribution

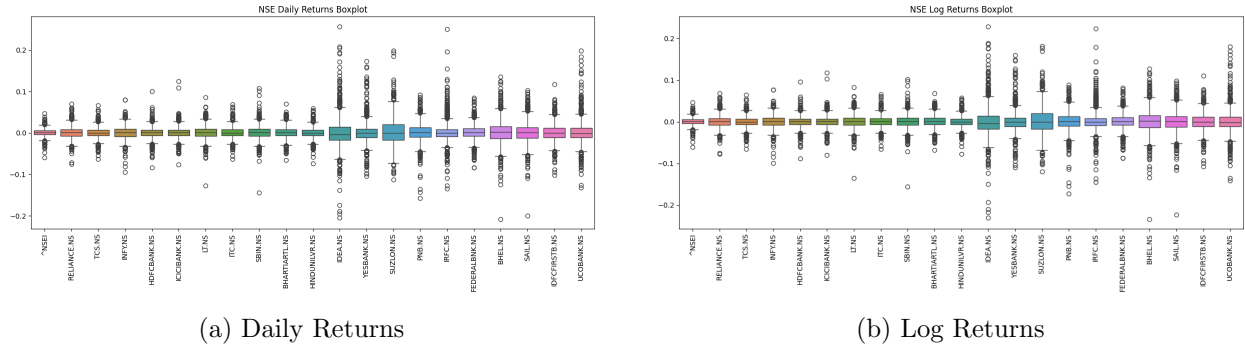
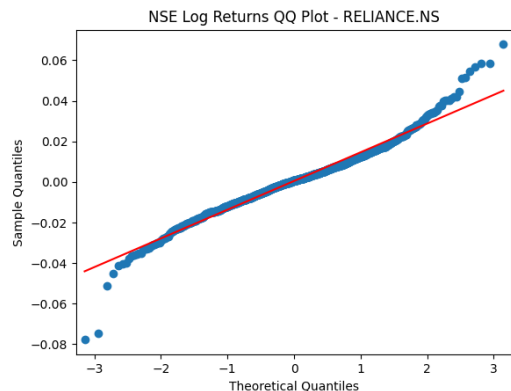
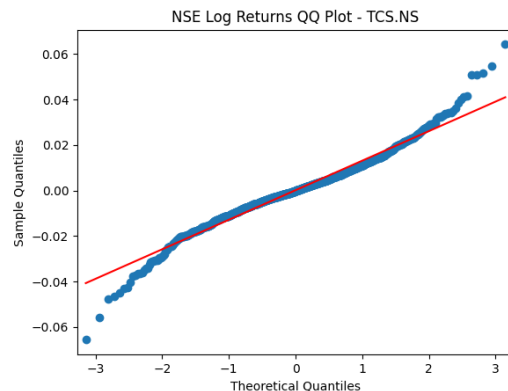


Figure 3: Comparison of daily and log return distributions

### 3.3 QQ Plots for Log Returns



(a) RELIANCE (NSE)



(b) TCS (NSE)

Figure 4: QQ plots of log returns

### 3.4 Normality Tests

|               | KS Statistic | p-value  |
|---------------|--------------|----------|
| $N_{SEI}$     | 0.056787     | 0.000758 |
| RELIANCE.NS   | 0.055180     | 0.001175 |
| TCS.NS        | 0.053320     | 0.001923 |
| INFY.NS       | 0.045423     | 0.012877 |
| HDFCBANK.NS   | 0.065788     | 0.000051 |
| ICICIBANK.NS  | 0.067329     | 0.000031 |
| LT.NS         | 0.056167     | 0.000899 |
| ITC.NS        | 0.058687     | 0.000444 |
| SBIN.NS       | 0.071215     | 0.000008 |
| BHARTIARTL.NS | 0.050993     | 0.003475 |
| HINDUNILVR.NS | 0.060611     | 0.000254 |
| IDEA.NS       | 0.105839     | 0.000000 |
| YESBANK.NS    | 0.113638     | 0.000000 |
| SUZLON.NS     | 0.070099     | 0.000012 |
| PNB.NS        | 0.069866     | 0.000013 |
| IRFC.NS       | 0.151526     | 0.000000 |
| FEDERALBNK.NS | 0.067293     | 0.000032 |
| BHEL.NS       | 0.061352     | 0.000203 |
| SAIL.NS       | 0.069866     | 0.000013 |
| IDFCFIRSTB.NS | 0.053075     | 0.002049 |
| UCOBANK.NS    | 0.105912     | 0.000000 |

|               | Shapiro Statistic | p-value  |
|---------------|-------------------|----------|
| $^N SEI$      | 0.957609          | 0.000000 |
| RELIANCE.NS   | 0.970289          | 0.000000 |
| TCS.NS        | 0.970915          | 0.000000 |
| INFY.NS       | 0.962087          | 0.000000 |
| HDFCBANK.NS   | 0.943990          | 0.000000 |
| ICICIBANK.NS  | 0.932836          | 0.000000 |
| LT.NS         | 0.950243          | 0.000000 |
| ITC.NS        | 0.962980          | 0.000000 |
| SBIN.NS       | 0.928223          | 0.000000 |
| BHARTIARTL.NS | 0.979730          | 0.000000 |
| HINDUNILVR.NS | 0.961578          | 0.000000 |
| IDEA.NS       | 0.899637          | 0.000000 |
| YESBANK.NS    | 0.876439          | 0.000000 |
| SUZLON.NS     | 0.969078          | 0.000000 |
| PNB.NS        | 0.934086          | 0.000000 |
| IRFC.NS       | 0.837534          | 0.000000 |
| FEDERALBNK.NS | 0.961160          | 0.000000 |
| BHEL.NS       | 0.946581          | 0.000000 |
| SAIL.NS       | 0.953918          | 0.000000 |
| IDFCFIRSTB.NS | 0.966583          | 0.000000 |
| UCOBANK.NS    | 0.881011          | 0.000000 |

### 3.5 MLE Estimates

|               | Mean (MLE) | Variance (MLE) | CI Lower  | CI Upper |
|---------------|------------|----------------|-----------|----------|
| $^N SEI$      | 0.000529   | 0.000077       | 0.000035  | 0.001024 |
| RELIANCE.NS   | 0.000503   | 0.000200       | -0.000292 | 0.001299 |
| TCS.NS        | 0.000126   | 0.000169       | -0.000604 | 0.000856 |
| INFY.NS       | 0.000326   | 0.000228       | -0.000522 | 0.001175 |
| HDFCBANK.NS   | 0.000336   | 0.000178       | -0.000414 | 0.001087 |
| ICICIBANK.NS  | 0.000782   | 0.000190       | 0.000006  | 0.001558 |
| LT.NS         | 0.000957   | 0.000232       | 0.000101  | 0.001813 |
| ITC.NS        | 0.000759   | 0.000156       | 0.000057  | 0.001462 |
| SBIN.NS       | 0.001086   | 0.000271       | 0.000160  | 0.002011 |
| BHARTIARTL.NS | 0.001131   | 0.000201       | 0.000335  | 0.001928 |
| HINDUNILVR.NS | 0.000078   | 0.000159       | -0.000632 | 0.000789 |
| IDEA.NS       | 0.000061   | 0.001373       | -0.002023 | 0.002144 |
| YESBANK.NS    | 0.000251   | 0.000602       | -0.001129 | 0.001631 |
| SUZLON.NS     | 0.001807   | 0.001078       | -0.000039 | 0.003654 |
| PNB.NS        | 0.001132   | 0.000530       | -0.000163 | 0.002427 |
| IRFC.NS       | 0.001471   | 0.000638       | 0.000051  | 0.002891 |
| FEDERALBNK.NS | 0.001114   | 0.000326       | 0.000099  | 0.002130 |
| BHEL.NS       | 0.001706   | 0.000784       | 0.000132  | 0.003281 |
| SAIL.NS       | 0.000871   | 0.000680       | -0.000595 | 0.002337 |
| IDFCFIRSTB.NS | 0.000485   | 0.000408       | -0.000651 | 0.001621 |
| UCOBANK.NS    | 0.000680   | 0.000726       | -0.000835 | 0.002195 |

## 4 Weekly Returns

Weekly returns are obtained by resampling daily prices and taking the final price of each week.

### 4.1 Descriptive Statistics

|               | Mean     | Std Dev  | Skewness  | Kurtosis  |
|---------------|----------|----------|-----------|-----------|
| $^N SEI$      | 0.002679 | 0.018698 | 0.116635  | 1.863309  |
| RELIANCE.NS   | 0.002820 | 0.029665 | 0.109721  | 0.351061  |
| TCS.NS        | 0.001039 | 0.029652 | -0.445196 | 0.373492  |
| INFY.NS       | 0.002095 | 0.033102 | -0.469137 | 0.326553  |
| HDFCBANK.NS   | 0.001970 | 0.027674 | 0.334388  | 3.575725  |
| ICICIBANK.NS  | 0.004106 | 0.028701 | 0.649263  | 1.695784  |
| LT.NS         | 0.005065 | 0.032729 | 0.157384  | 0.548766  |
| ITC.NS        | 0.004033 | 0.029846 | 0.790512  | 3.052070  |
| SBIN.NS       | 0.005879 | 0.039983 | 3.590572  | 33.141437 |
| BHARTIARTL.NS | 0.005771 | 0.028722 | 0.141980  | 0.191911  |
| HINDUNILVR.NS | 0.000781 | 0.028826 | 0.505730  | 1.240541  |
| IDEA.NS       | 0.003323 | 0.079379 | 0.804717  | 2.110417  |
| YESBANK.NS    | 0.002694 | 0.056311 | 1.474331  | 5.639910  |
| SUZLON.NS     | 0.011559 | 0.079249 | 0.908999  | 1.264966  |
| PNB.NS        | 0.006528 | 0.048892 | 0.651855  | 2.139491  |
| IRFC.NS       | 0.008923 | 0.065576 | 2.446570  | 10.424914 |
| FEDERALBNK.NS | 0.006035 | 0.039170 | 0.421376  | 1.162501  |
| BHEL.NS       | 0.010097 | 0.064245 | 0.609561  | 1.610508  |
| SAIL.NS       | 0.005632 | 0.055783 | 0.810750  | 2.655032  |
| IDFCFIRSTB.NS | 0.003188 | 0.042575 | 0.464756  | 2.202099  |
| UCOBANK.NS    | 0.005143 | 0.065267 | 2.858284  | 19.449737 |

### 4.2 Weekly Return Boxplots

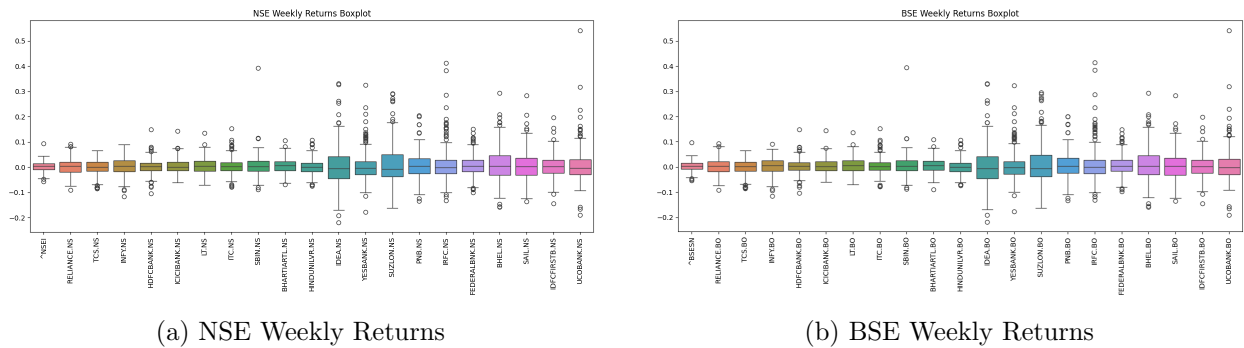


Figure 5: Weekly return distributions

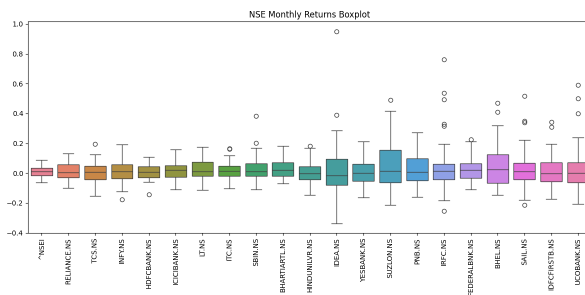
## 5 Monthly Returns

Monthly returns are obtained by selecting the final trading price of each month.

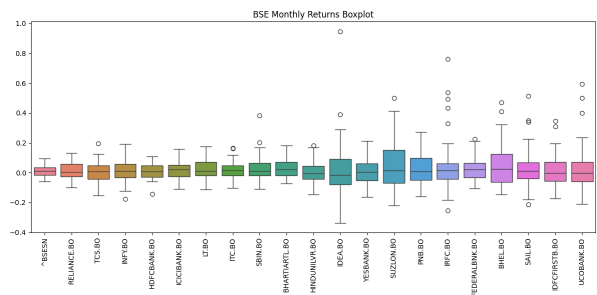
## 5.1 Descriptive Statistics

|               | Mean     | Std Dev  | Skewness  | Kurtosis  |
|---------------|----------|----------|-----------|-----------|
| $N_{SEI}$     | 0.011588 | 0.035974 | 0.106941  | -0.547111 |
| RELIANCE.NS   | 0.012104 | 0.059013 | 0.224393  | -0.874196 |
| TCS.NS        | 0.004461 | 0.061647 | 0.183146  | 0.613054  |
| INFY.NS       | 0.008976 | 0.067375 | -0.065216 | 0.391298  |
| HDFCBANK.NS   | 0.008127 | 0.048938 | -0.148332 | 0.291098  |
| ICICIBANK.NS  | 0.017639 | 0.054292 | 0.306279  | 0.027467  |
| LT.NS         | 0.021551 | 0.058793 | 0.380618  | -0.365183 |
| ITC.NS        | 0.017169 | 0.054315 | 0.497962  | 0.440235  |
| SBIN.NS       | 0.025776 | 0.083858 | 1.433364  | 4.126494  |
| BHARTIARTL.NS | 0.025083 | 0.056378 | 0.332199  | -0.492050 |
| HINDUNILVR.NS | 0.003687 | 0.065468 | 0.461007  | 0.319464  |
| IDEA.NS       | 0.016345 | 0.189377 | 2.000361  | 8.464091  |
| YESBANK.NS    | 0.009180 | 0.091330 | 0.420033  | -0.325933 |
| SUZLON.NS     | 0.050272 | 0.165985 | 0.636800  | -0.207084 |
| PNB.NS        | 0.028597 | 0.103568 | 0.449374  | -0.562615 |
| IRFC.NS       | 0.041684 | 0.165498 | 2.200919  | 6.370001  |
| FEDERALBNK.NS | 0.025806 | 0.074594 | 0.695910  | 0.318178  |
| BHEL.NS       | 0.043812 | 0.134260 | 0.875022  | 0.906858  |
| SAIL.NS       | 0.024642 | 0.121923 | 1.508532  | 4.120308  |
| IDFCFIRSTB.NS | 0.014891 | 0.102446 | 0.928278  | 1.214926  |
| UCOBANK.NS    | 0.023177 | 0.146550 | 1.803935  | 4.232154  |

## 5.2 Monthly Return Boxplots



(a) NSE Monthly Returns



(b) BSE Monthly Returns

Figure 6: Monthly return distributions

## 6 Conclusion

The empirical analysis reveals several well-known stylized facts of financial time series:

- Stock returns exhibit heavy tails and extreme observations.
- Normality is rejected by both Kolmogorov–Smirnov and Shapiro–Wilk tests.

- Log returns behave similarly to simple returns but are theoretically consistent with continuous-time models.
- Aggregating returns over longer horizons increases the variance of returns.

These findings highlight the limitations of the normality assumption used in classical financial models such as the Black–Scholes framework.